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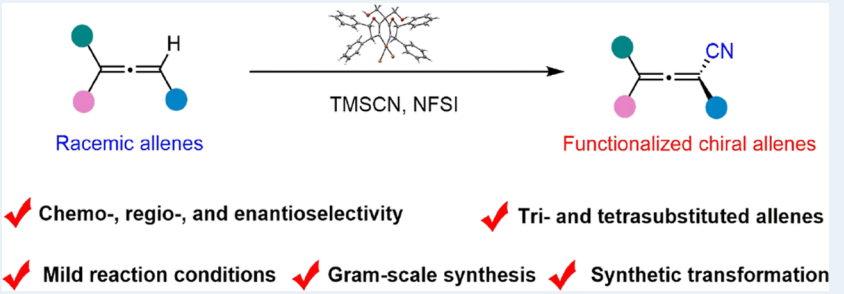
Copper-Catalyzed Enantioselective Radical Cyanation of Allenic C(sp^2)−H Bonds to Access Tri- and Tetrasubstituted Chiral Allenes

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ABSTRACT: The asymmetric radical functionalization of allenic C(sp^2)−H bonds represents an appealing strategy for synthesizing axially chiral allenes. This method enables us to generate the more structurally complex and highly functionalized axially chiral allenes directly by using the readily accessible and less substituted racemic allenes while concurrently achieving apparent dynamic kinetic resolution. However, compared to alkyne derivative-based approaches, the asymmetric functionalization of allenic C(sp^2)−H bonds remains a challenge primarily due to the intertwined demands of chemo-, regio-, and enantioselectivity. Here, we report a copper-catalyzed asymmetric cyanation of allenic C(sp^2)−H bonds by using a substituted oxazoline chiral ligand. This strategy efficiently transforms low-substituted racemic allenes into highly functionalized, axially chiral allenyl nitriles in a highly chemo-, regio-, and enantioselective manner.

KEYWORDS: *enantioselective cyanation, C−H functionalization, radical, chiral allene, copper catalysis*

INTRODUCTION

Axially chiral allenes are a class of significant organic molecules, characterized by their distinctive orthogonal twisted π-systems.[1] Functionalized allenes often exhibit notable physiological activities, such as enzyme inhibition, cytotoxicity, and antibacterial properties (Scheme 1a).[2] As a result, structural modifications of allene compounds or their incorporation into pharmaceutical molecules to enhance biological activity represent a key direction for future development in this field.[3]

Over the past decade, significant progress has been made in the development of radical synthesis methods for allenes.[4] Among these advancements, the asymmetric transformation of allenyl radicals has emerged as a powerful strategy for constructing axially chiral allenes (Scheme 1b, left). Our group previously reported the first copper-catalyzed radical 1,4-bifunctionalization of 1,3-enynes, achieving asymmetric 1,4-oxycyanation and offering a novel method for synthesizing elusive axially chiral allenes.[5] Subsequently, Deng et al. demonstrated a stereoselective decarboxylation strategy for 1,3-enynes using a photoinduced dual catalytic system.[6] Meanwhile, Liu[7] and Guo[8] independently achieved copper-catalyzed asymmetric 1,4-alkylalkynylation and arylalkynylation via allenyl radical-terminal alkyne coupling. Notably, He et al. introduced a nickel-catalyzed radical cascade reaction for enantio- and regioselective 1,4-arylalkylation of 1,3-enynes.[9] These enantioselective radical 1,4-difunctionalization protocols collectively provide efficient access to structurally diverse, enantioenriched tetrasubstituted allenes. Expanding the scope further, Liu et al. reported copper-catalyzed enantioselective radical cyanation of propargylic C(sp^3)−H bonds, delivering chiral trisubstituted allenyl nitriles.[10] These outstanding works demonstrate that constructing axially chiral allenes from alkyne derivatives, such as 1,3-enynes or alkynes, has become a reliable and promising strategy.

From another perspective, the asymmetric radical functionalization of allenic C(sp^2)−H bonds also represents an attractive strategy for synthesizing axially chiral allenes (Scheme 1b, right). This approach enables the functionalization of readily available low-substituted racemic allenes while simultaneously achieving apparent dynamic kinetic resolution. As a result, this provides access to structurally complex and

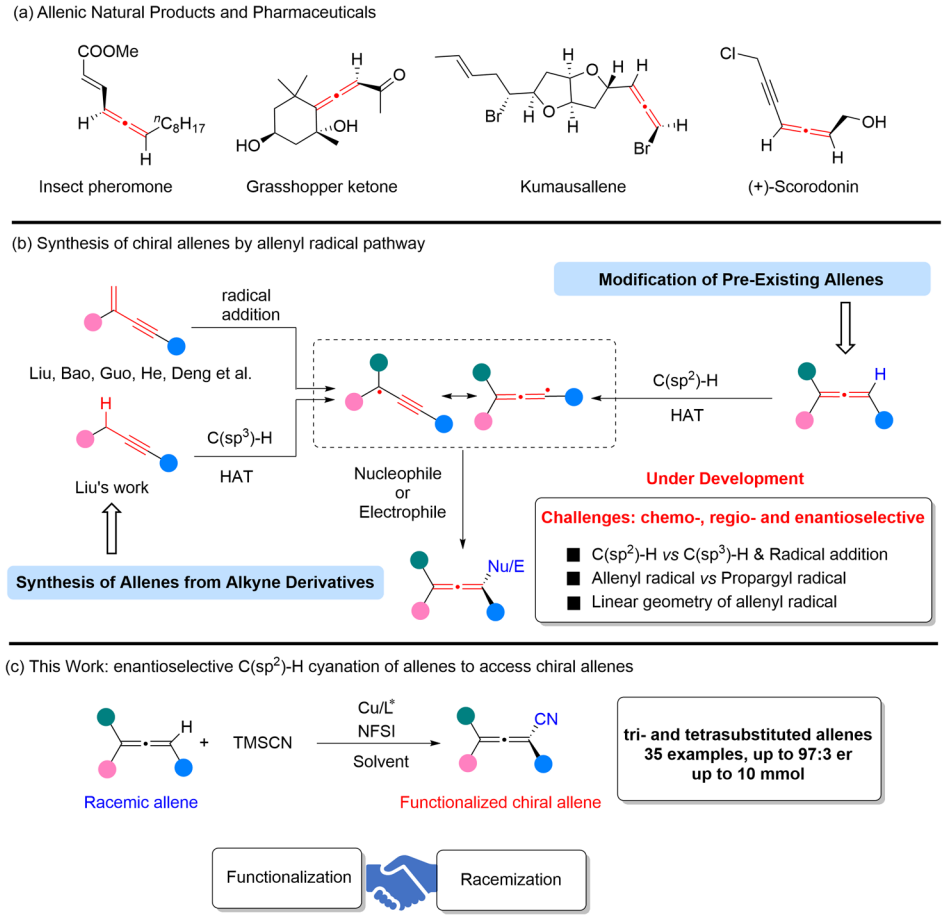
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Scheme 1. Enantiocontrol of Allenyl Radicals to Access Chiral Allenes



highly functionalized axially chiral allenes. However, this strategy faces significant challenges, particularly in achieving satisfactory stereoselective control and the addressing issues of chemical and regioselectivity during the activation of allenic C(sp^2)−H bonds and the transformation of allenyl radicals. In 2016, Lee and colleagues demonstrated the use of peroxyl radicals to abstract hydrogen atoms from allenic C−H bonds, thereby generating allenyl radical intermediates.[11] More recently, Liu[12] and Wu[13] almost simultaneously introduced a groundbreaking strategy for the functionalization of allenes via radical activation of allylic C(sp^2)−H bonds. These elegant studies achieved high chemoselectivity and regioselectivity, paving the way for the synthesis of complex and functionalized allenyl derivatives from simple precursors. Despite these advances, the asymmetric radical functionalization of allenic C(sp^2)−H bonds remains a significant challenge due to the intertwined demands of chemo- [C(sp^2)−H vs C(sp^3)−H activation vs radical addition], regio- (allenyl vs propargyl radical pathways), and enantioselectivity (the linear geometry of allenyl radicals).[12a,13a]

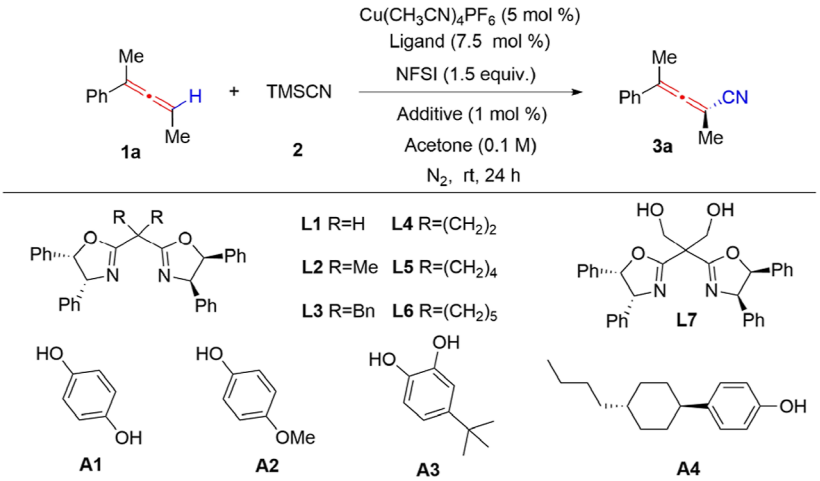
In this context, we have conducted extensive studies to address these challenges, and herein, we report an asymmetric cyanation reaction of C(sp^2)−H bonds catalyzed by a copper complex with a substituted oxazoline (Scheme 1c). This approach enables the efficient transformation of low-substituted racemic allenes into highly functionalized, axially chiral allenyl nitriles under mild conditions. This work not only addresses the challenges of chemo-, regio-, and enantioselectivity but also highlights the potential of C−H functionalization strategies for the synthesis of complex organic molecules.

METHODS

Reaction Optimization. To initiate these studies, (1-methyl-1,2-butadienyl)benzene (**1a**) was reacted as the model substrate, with *N*-fluorobenzenesulfonimide (NFSI) as the HAT reagent and trimethylsilyl cyanide (**2**) as the coupling partner in the presence of a catalytic amount of Cu-$(CH_3CN)_4PF_6$ (5 mol %) in acetone at room temperature for 24 h. To our delight, the desired product **3a** was obtained in 25% yield and 64:36 er when bis(oxazoline) **L1** (7.5 mol %) was used as the ligand (Table 1, entry 1). We then screened a series of bis(oxazoline) ligands (**L2**−**L7**) with different substituents on R, finding that **L7** gave the best result (25% yield, 91:9 er) (entry 7). At 0 °C, both the yield and er were improved compared to room temperature (entry 8). Given the propensity of the reaction substrates to polymerize, a number of inhibitors were tested as additives.[14] It was found that the addition of 1 mol % of 4-(*trans*-4-butylcyclohexyl)phenol (**A4**) significantly increased the reaction yield without compromising the high enantioselectivity. Finally, examination of different copper salts showed that $CuBr_2$ was the best choice, yielding 58% yield, 93:7 er (for details of optimization of the conditions, see Supporting Information). Therefore, the optimal conditions were established as $CuBr_2$/**L7** (1:1.5), NFSI (1.5 equiv), TMSCN (3 equiv), **A4** (1 mol %), 0 °C, 48 h.

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Table 1. Optimization of the Reaction Conditions[a,b,c]



entry	ligand	additive	yield (%)	Er
1	**L1**		25	64:36
2	**L2**		22	80:20
3	**L3**		21	81:19
4	**L4**		26	80:20
5	**L5**		24	75:25
6	**L6**		21	80:20
7	**L7**		25	91:9
8[d]	**L7**		35	93:7
9[d]	**L7**	**A1**	21	90:10
10[d]	**L7**	**A2**	26	87:13
11[d]	**L7**	**A3**	50	92:8
12[d]	**L7**	**A4**	50	93:7
13[d,e]	**L7**	**A4**	58	93:7

[a]Reaction conditions: Cu catalyst (5 mol %), ligand (7.5 mol %), **1a** (0.1 mmol), NFSI (0.15 mmol), TMSCN (0.3 mmol), in acetone under N_2 at rt for 24 h. [b]Isolated yield. [c]Er values determined by HPLC on a chiral stationary phase. [d]0 °C for 48 h. [e]$CuBr_2$ as the catalyst.

Substrate Scope. Having established the optimized reaction conditions, we tested a variety of substituted allenes to probe the versatility of our catalytic system (Scheme 2). A variety of enantioenriched tetrasubstituted allenyl nitriles (**3a**−**3y**) were obtained in moderate to good yields (43−62%) and with high enantiomeric ratios (82:18−97:3 er). Allene substrates with electron-donating or electron-withdrawing groups introduced at various positions on the aromatic ring can provide the corresponding products (**3a**−**3h**) with an er value as high as 97:3. Moreover, other aromatic rings such as the naphthyl group (**3i** and **3j**) and thienyl group (**3k**) are tolerated. A screening of the R^1 and R^2 revealed that alkyl chains (**3l**−**3q**), phosphonate group (**3r**), silyl ether group (**3s**), chloroalkyl group (**3t**), benzyl group (**3u**), and vinyl groups (**3v** and **3w**) are tolerated. Notably, a diverse array of substrates, encompassing those with benzyl C−H bonds (**3u**) as well as allyl C−H bonds (**3v** and **3w**), have been selectively transformed through the cyanation of the allenic C(sp^2)−H bond, affording the target products without the observation of C(sp^3)−H bond cyanation. The absolute configuration of axially chiral allenyl nitriles **3c** was unambiguously confirmed by X-ray analysis. Both **3x**[12a] and **3y**[13a] can also be obtained in good yields as well as with higher enantioselectivities by our method. To showcase the practical application of our method, the allenic C(sp^2)−H cyanation reaction was scaled up to a 10 mmol batch. This amplification synthesis exhibits even higher enantioselectivity compared to standard conditions (**3a**, 96:4 er and **3y**, 93:7 er), highlighting the enormous synthetic potential of our method.

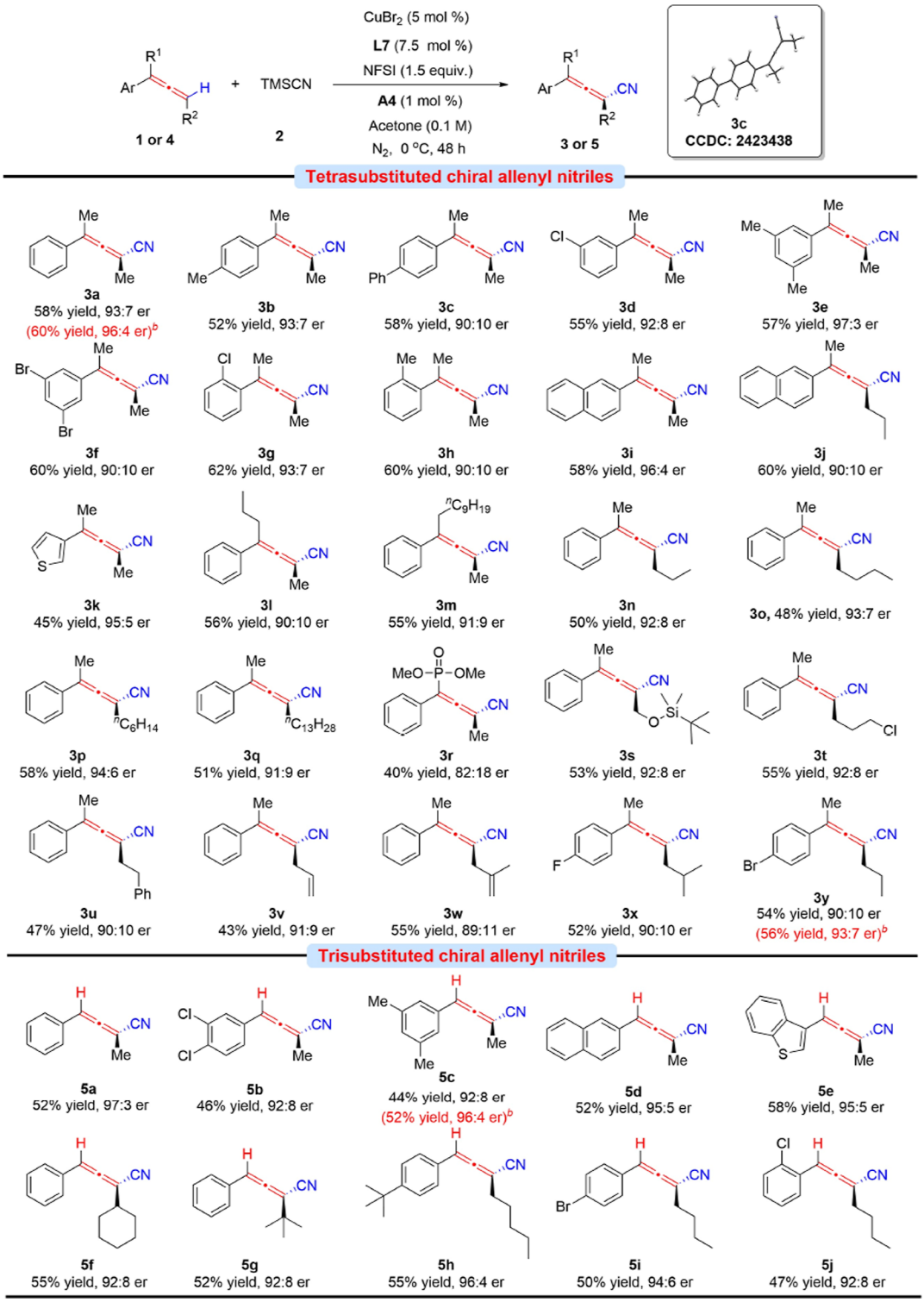
The versatility of this strategy extends to the synthesis of trisubstituted allenyl nitriles. Under the standard reaction conditions, a spectrum of enantioenriched trisubstituted allenyl nitriles (**5a**−**5j**) was successfully synthesized. The process demonstrates the well tolerance for diverse functional groups, including phenyl (**5a**−**5c**), naphthyl (**5d**), benzothiophene (**5e**), and secondary and tertiary alkyl groups (**5f** and **5g**). Products **5h**−**5j** can also be obtained in good yields as well as with higher enantioselectivities by our method.[12a,13a] Moreover, scaling up the reaction has led to a further enhancement in enantioselectivity, as evidenced by product **5c**, which achieved an enantiomeric ratio of 96:4.

Further Transformations. Such obtained enantioenriched allenyl nitriles potentially have wide applications in organic synthesis (Scheme 3). For example, the reduction of allenyl nitrile **3a** with 1.1 equiv of DIBAL-H efficiently yields allenyl aldehyde **6**,[15] which can be further reduced to produce allenyl propanol **7**. Additionally, chiral allenyl nitrile **3a** can be readily transformed into allenyl amide **8** in a high yield through hydrolysis. In the presence of potassium hydroxide, compound **3a** can be fully hydrolyzed to generate the corresponding acid **9**.[15]

The construction of chiral quaternary carbon centers remains a major challenge in synthetic chemistry, and the development of methods to form such chiral centers is an important goal. In this context, the transfer of axial chirality to

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Scheme 2. Scope of Chiral Allenyl Nitriles[a]



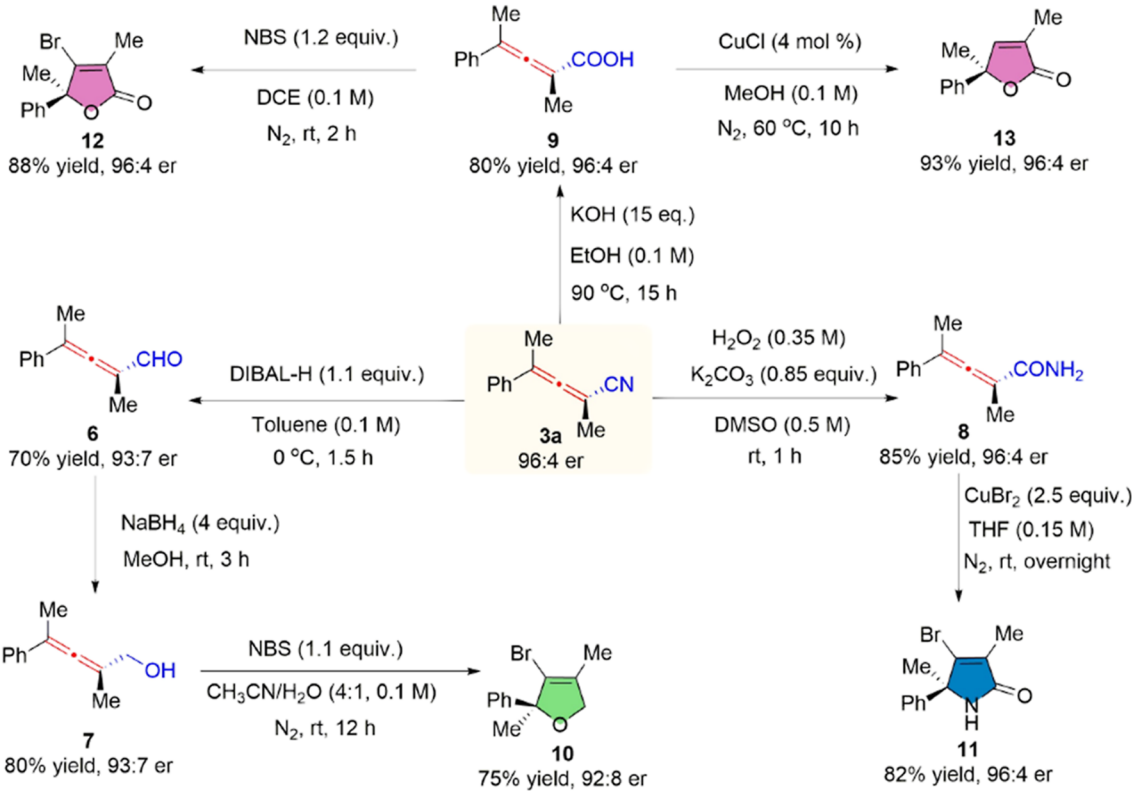
[a]Reactions were conducted on a 0.2 mmol scale with the reaction condition of entry 13 in Table 1. [b]10 mmol scale.

central chirality was explored. For instance, treatment of **7** with NBS leads to the formation of enantioenriched cyclic ether **10** with a chiral quaternary carbon center.[8] Furthermore, chiral allenyl amide **8** reacts smoothly with $CuBr_2$ to produce the highly functionalized lactam **11**.[16] Cyclization of allenyl acid **9** effectively generates chiral butenolides **12** and **13**, which are crucial structural motifs for numerous biologically active compounds.[17] Notably, the enantioselectivity was retained in all of these reactions.

Mechanistic Investigations. To elucidate the reaction mechanism, we conducted a series of experiments (Scheme 4a–f). Initial radical scavenger studies using 2,2,6,6-tetramethylpiperidine-1-oxyl or butylated hydroxytoluene under standard conditions completely suppressed product formation,

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Scheme 3. Further Transformations of Products



confirming a radical-mediated pathway (Scheme 4a). A radical-clock experiment with an alkenyl cyclopropane substrate generated the ring-opening product (**15**), directly evidencing the allenyl radical intermediate (Scheme 4b). Kinetic isotope effect studies (k_H/k_D = 2.3) suggest that intermolecular hydrogen atom abstraction might be the rate-limiting step (Scheme 4c).

We then proceeded to investigate the nonlinear effects in this method and found that allenyl nitrile **3a** indeed exhibited linear effects (Scheme 4d).[18] This led us to consider that a monoligand copper complex acts as the active catalytic species in the reaction.[19] To delve deeper into the structure of the metal complex, we conducted control experiments under standard conditions using ligand **L8**, with its hydroxyl group protected by trimethylsilyl; these results closely mirrored those obtained under standard conditions (Scheme 4e, 58% yield, 93:7 er vs 50% yield, 93:7 er). This finding implies that the hydroxyl group as an electron-donating group may be a spectator and the hydrogen bonding may not play a role in the metal complex. Furthermore, the structure of **L7**$CuBr_2$ (**16**) was elucidated through X-ray crystallography, revealing the complex composed of a single copper atom and a single ligand (Scheme 4f). Upon subjecting this crystal to a cyanation reaction as the catalyst, we observed that the reaction efficiencies and enantioselectivities were nearly identical to those under standard conditions (47% yield, 92:8 er), thereby validating the existence of a monoligand copper active species. This is also consistent with the conclusions drawn from the linear effect observed earlier.

Drawing from these experimental outcomes, we propose a potential catalytic cycle mechanism (Scheme 5). Initially, a monovalent monoligand copper species **A** undergoes single-electron transfer with NFSI, leading to the formation of a divalent fluorinated copper species **B** and a nitrogen radical. This nitrogen radical can abstract the C(sp^2)−H bond from the allene, forming an allenyl radical.[12a,13a] The divalent fluorinated copper species **B** then undergoes transmetalation with TMSCN, yielding cyanated copper intermediate **C**. Ultimately, intermediate **C** engages in an asymmetric functional group transfer with the allenyl radical, yielding the target product and regenerating the catalyst.[20]

CONCLUSIONS

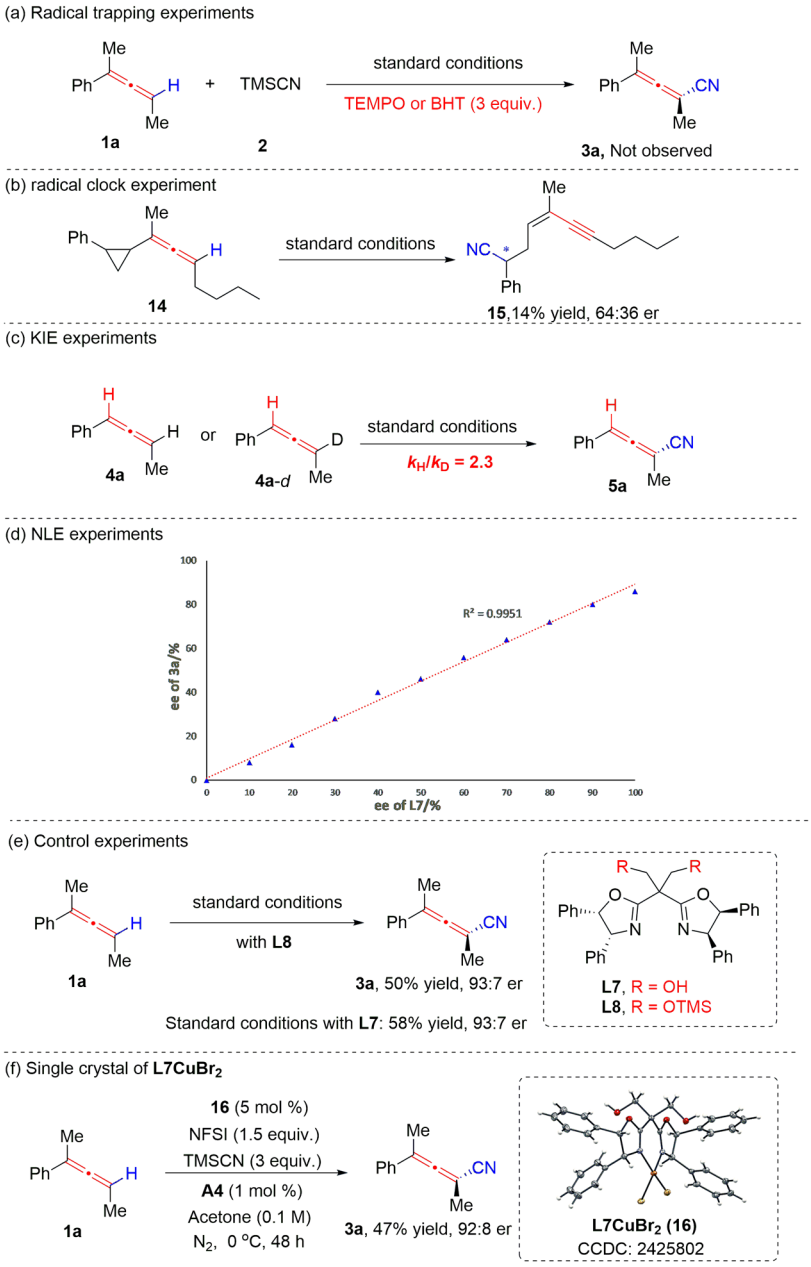
In conclusion, we have developed a copper-catalyzed enantioselective radical cyanation of allenic C(sp^2)−H bonds under mild reaction conditions, enabling efficient access to trisubstituted and tetrasubstituted axially chiral allenyl nitriles. This strategy achieves high chemoselectivity, regioselectivity, and enantioselectivity, with broad functional group tolerance. Mechanistic studies corroborate a radical pathway involving a monoligand copper species. The synthetic utility of this method is further highlighted by its scalability (10 mmol scale) and the versatile downstream derivatization with a chiral quaternary carbon center of chiral allenyl nitriles into cyclic ether, lactam, and butenolides. We believe that this work may open up new possibilities of allenic C(sp^2)−H asymmetric functionalization, offering a promising entry to highly functionalized axially chiral allene derivatives.

MATERIALS AND METHODS

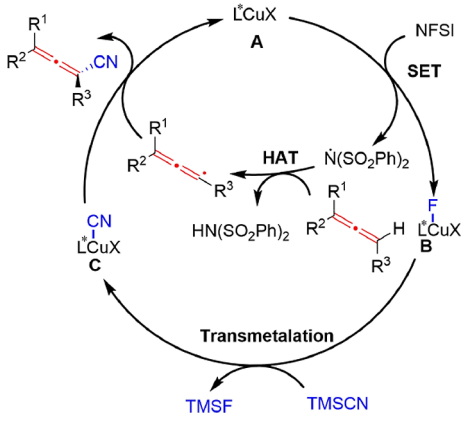
All information pertaining to the materials and methods used in this study is provided in the Supporting Information.

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Scheme 4. Mechanistic Investigations



Scheme 5. Proposed Reaction Mechanism



■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c01796.

Experimental procedures; properties of some compounds; copies of ^{1}H and ^{13}C spectra of the products; and cif files for solid-state structures (PDF)

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Notes

The authors declare no competing financial interest.

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